# MARCELO CARVALHO FILHO

SITE RELIABILITY ENGINEER | KUBERNETES · AWS · TERRAFORM · GITOPS

Palm Bay, Florida, USA | Relocating to Spain, summer 2027 (employer sponsorship required)
marcelo.carvalho.sre@gmail.com | +1 (321) 387-2446
jmarcelocarvalho.com | linkedin.com/in/jmarcelo-carvalho | github.com/jmarcelo81

## CAREER HIGHLIGHTS

**Reliability at Scale:** Technical escalation lead for 24/7 incident response across a 1,200+ server multi-tenant Linux/Windows fleet; root-cause analysis, vulnerability remediation and runbook standardization helped cut monthly outages from about 10 to 3–4.

**Automation:** Automated the WHM/cPanel CentOS 7 to Ubuntu 22.04 / AlmaLinux modernization across 200+ servers with Ansible, cutting manual intervention by 60% and eliminating configuration drift.

**Migrations:** Led 100+ production website migrations with controlled 2-hour cutovers; Jenkins and Ansible automation cut average deployment time for new managed customer environments from 18.8 to 5.6 days.

**Platform Engineering:** Run a 4-node K3s cluster with ArgoCD GitOps and a 3-node OpenStack private cloud in a self-hosted lab; Terraform and GitHub Actions deploy my production portfolio site on AWS.

**Credentials:** Almost 4 years of full-time infrastructure experience, built on earlier IT consulting and judicial operations management; AWS Solutions Architect – Associate, HashiCorp Terraform Associate, CompTIA Security+ and Cloud+; Portuguese, English and Spanish (C1).

## AREAS OF EXPERTISE

- Incident Response
- Root-Cause Analysis
- AWS (EC2, S3, VPC, IAM, RDS, CloudWatch)
- Linux & Windows Server
- HAProxy, Nginx & Load Balancing
- Kubernetes (K3s) & ArgoCD GitOps
- Ansible, Python & Bash Automation
- Prometheus, Grafana & Alertmanager
- HIPAA / PCI-DSS Secure Design
- SIEM & IPS/IDS
- Python
- Terraform & Infrastructure as Code
- CI/CD (GitHub Actions, Jenkins)
- OpenStack, KVM & VMware vSphere
- PostgreSQL, MySQL, MSSQL & Redis

*Kubernetes, ArgoCD and OpenStack: self-hosted lab. Terraform and GitHub Actions: personal production site. All other tools, including Jenkins and Ansible: used at Atlantic.Net.*

## PROFESSIONAL EXPERIENCE

**ATLANTIC.NET, Orlando, FL (hybrid)** 12/2022 – Present

*Continuous employment; part-time schedule 03/2025 – 11/2025 during the concurrent GE Vernova contract below.*

**DATA CENTER MSP TECHNICAL SUPPORT LEAD** 07/2025 – Present

- **Incident Command:** Technical escalation lead for 24/7 incident response across a 1,200+ server multi-tenant Linux/Windows fleet; root-cause analysis and vulnerability remediation contributed to cutting monthly outages from 10 to 3–4 (ticketing-system data).
- **Recovery Time:** Cut mean time to recovery (MTTR) 25% on average by standardizing 15+ run books and escalation paths.
- **Security Response:** Lead response to actively exploited malware and vulnerability campaigns across customer servers including the recent PHP2Shell exploitation/issues - CVE-2026-109618 and CVE-2026-109619.
- **Team Development:** Mentor technicians (team of 4, up to 9) on troubleshooting, hardening and on-call readiness.
- **Secure Architecture:** Design HIPAA/PCI-DSS-compliant environments (OPNsense, FortiGate, CrowdSec, IPS/IDS, SIEM) and multi-domain HAProxy load-balancing tiers over Apache/Nginx/IIS with MSSQL, MySQL and PostgreSQL backends.
- **Deployment Automation:** Built Jenkins and Ansible playbooks that deploy VMs on the KVM infrastructure, create users, generated strong passwords, install standard administrator packages, apply SSH hardening and create nftables rulesets; average deployment time for new managed customer environments fell from 18.8 to 5.6 days.
- **Migrations & Platform:** Directed lift-and-shift and platform migrations; operate the KVM/virsh hosting platform (thousands of VMs) with YAML-based provisioning and capacity planning.

**ESCALATIONS SPECIALIST** 09/2023 – 07/2025

- **Escalation:** Final escalation point for high-impact incidents across web, database and hybrid-cloud environments (Nginx/Apache, PHP, Python; PostgreSQL, MySQL, MariaDB), restoring service within strict SLAs; wrote runbooks and ran postmortems that produced corrective actions.
- **Network Isolation:** Traced cluster-wide broadcast storms to an ebtables misconfiguration on hypervisor nodes and restored tenant traffic isolation, cutting customer-reported performance issues by 50% ((ticketing-system data).
- **Ansible Modernization:** Automated the CentOS 7 to Ubuntu 22.04 / Alma Linux migration across 200+ servers, cutting manual intervention 60% and eliminating configuration drift.
- **Site Migrations:** Led migrations of 100+ production websites (cPanel transfers, MySQL replication, controlled 2-hour cutovers) plus cloud-to-cloud and on-premises to cloud moves, validating performance and security afterwards.

**NOC HELP DESK TIER 2** 12/2022 – 09/2023

- **Operations:** Monitored a 25,000 sq ft data center and 7 remote hosting sites for 200+ clients; resolved about 400-500 tickets a month, including work with Tier-3 ISPs.
- **Security & Networking:** Deployed Wazuh SIEM, pfSense/OPNsense firewalls, cPanel, Plesk and NextCloud; helped upgrade 300+ customer firewalls; managed load balancing, CARP and IPsec tunnels.
- **Vulnerability Management:** Researched and remediated vulnerabilities across more than 70% of client environments and managed HIPAA and PCI-compliant environments to customer requirements.

**SOGETI, CONTRACT AT GE VERNOVA, Remote (Melbourne, FL)** 03/2025 – 11/2025

**SYSTEM ADMINISTRATOR, PATCH MANAGEMENT** *(concurrent with part-time work at Atlantic.Net)*

- **Patch Automation:** Engineered Python, Bash and PowerShell patch-lifecycle automation for 50+ production VMs (Red Hat 8, Oracle Linux 7, Windows Server) supporting national power-grid EMS/SCADA systems, cutting maintenance windows 30% and raising patch compliance to 95%+ with zero unplanned downtime.
- **Virtualization & Compliance:** Ran a VMware patch-test lab, vSphere/ESXi capacity planning and zero-downtime migrations; used Veeam for recovery and CIS benchmark audits with the cybersecurity team; deployed fixes to live, air-gapped systems with minimal disruption.

## EARLIER EXPERIENCE

**JUDICIAL LAW CLERK MANAGER** *State of São Paulo Judiciary* 08/2015 – 07/2022

- Led a team of 4 clerks and 2 interns producing about 720 rulings a month; 100+ research projects cut average lawsuit resolution time by 41% (from 17 to 10 months). Coordinated a Center for Conflict Resolution for 18 months (90% settlement rate; active lawsuits down from 14,000 to 11,000).

**IT CONSULTANT** *Vendrameto Advogados & Aspen Chocolates (concurrent with judicial career)* 02/2012 – 06/2022

- Designed network segmentation (VLANs, firewall) and least-privilege Active Directory/GPO access for law firms and a manufacturer; built and supported Windows Server infrastructure (Active Directory, file and print servers).

## PRODUCTION & LAB PROJECTS

- **Production portfolio site on AWS (jmarcelocarvalho.com):** Dockerized Flask site behind Nginx on EC2 with PostgreSQL, Terraform-managed infrastructure, a GitHub Actions CI/CD pipeline, Let's Encrypt TLS and automated S3 backups.
- **Kubernetes SRE lab (github.com/jmarcelo81/kubernetes_lab):** 4-node K3s cluster managed with Ansible and deployed through ArgoCD GitOps, with Traefik ingress, cert-manager, Vault, Prometheus/Grafana/Alertmanager and PatchMon.
- **Private cloud lab:** 3-node OpenStack on Dell PowerEdge servers with Ceph RBD, Neutron over VxLAN, Heat, RGW-Swift and Wazuh monitoring.
- **Infrastructure as code (github.com/jmarcelo81/terraform-multi-cloud):** Terraform provisioning on both AWS and OpenStack.

## CERTIFICATIONS

**Cloud & IaC:** AWS Certified Solutions Architect – Associate (Jun 2026 – Jun 2029) · HashiCorp Certified: Terraform Associate 004 (Apr 2026 – Apr 2028) · AWS Certified Cloud Practitioner (Feb 2026 – Feb 2029) · CompTIA Cloud+ ce (Dec 2025 – Dec 2028) · CompTIA Secure Cloud Professional, CSCP (Dec 2025 – Aug 2028) · AWS Academy Cloud Architecting (Aug 2025)

**Security & Foundations:** CompTIA Security+ ce (Aug 2023 – Aug 2029) · CompTIA A+ ce (Oct 2022 – Aug 2029) · Microsoft SC-900 (Feb 2024) · Fortinet NSE 3, FortiGate 7.6 Operator (Apr 2026)

**In Progress:** AWS Certified AI Practitioner

## EDUCATION & LANGUAGES

**B.Sc. Computer Science:** Faculdade Anhanguera, São Paulo, Brazil | Online program, completed alongside full-time work (in progress, expected December 2028)

**Bachelor of Laws (LL.B.):** Faculdades Integradas de Itapetininga (FII-FKB), Brazil

**Languages:** Portuguese (native) · English (fluent) · Spanish (C1) · Italian (B2)